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E-205 MOTOR CONTROL CENTERS

1. SCOPE OF WORK

The Work shall include the design, supply, factory testing, delivery to the Site, installation, testing and commissioning of all motor control centers as part of the Work in accordance with Part 4, Subsection 1.2 Summary of Work.

Spare complete feeder shall be provided at least 10% for motor and outgoing feeders.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards, and the following specific standards.

2.1 National

PUIL	:	Peraturan Umum Instalasi Listrik - Indonesia (General Regulation of Electrical Installation Indonesia)
SPLN	:	Standard Perusahaan Umum Listrik Negara

2.2 International Electrotechnical Commission

IEC 60028	:	International standard of resistance for copper
IEC 60044	:	Instrument transformers
IEC 60186	:	Voltage transformers
IEC 60269	:	Low-voltage fuses
IEC 60277	:	Definitions for switchgear and controlgear
IEC 60439	:	Low-voltage switchgear and controlgear assemblies
IEC 60445	:	Identification of apparatus terminals and general rules for a uniform system of terminal marking, using an alphanumeric notation
IEC 60947	:	Low-voltage switchgear and control gear

2.3 American National Standards Institute

ANSI C37.20 : Standard for Switchgear Assemblies and Metal-Enclosed Bus

3. OPERATING AND DESIGN CONDITIONS

3.1 General

All MCCs shall be suitable for operation in a combined cycle power plant located in a tropical climate along the seacoast. Refer to Part 4, Subsections 1.1, 2.14 and 2.16 for the meteorological, seismic and environmental conditions at the Site.

3.2 Equipment Ratings

3.2.1 Circuit Breaker and Contactor Ratings

(1)	Number of phases	:	3
(2)	Nominal system voltage	:	400 V AC.
(3)	Rated voltage	:	440 V AC.
(4)	Rated frequency	:	50 Hz
(5)	Rated normal current		
	a. Main breakers	:	as required *
	b. Feeder breakers	:	as required*
(6)	Rated short-circuit breaking current (rms)	:	50 kA*
(7)	Rated short-circuit making current (peak)	:	125 kA*
(8)	Short circuit performance category	:	P-2
(9)	System grounding	:	Solid
(10)	Starter control circuit voltage	:	110 V AC

Note: *) Item to be determined by the Contactor and subject to approval by the Owner.

3.2.2 Bus Ratings

(1)	Continuous current		
	a. Main bus	:	*
	b. Vertical bus	:	*

- (2) Momentary current : 125 kA*
- (3) Temperature rise at rated continuous current above outside ambient of 40°C shall be in accordance with IEC Standard.

Note: *) Item to be determined by the Contactor and subject to approval by the Owner.

3.2.3 Transfer Switches

- a. Number of phases : 3
- b. Nominal system voltage : 400 V AC
- c. Rated voltage : 440 V AC
- d. Rated frequency : 50 Hz
- e. Continuous rating : 800 A*
- f. Momentary rating : 105 kA*

Note: *) Item to be determined by the Contactors and subject to approval by the Owner.

4. DETAILED REQUIREMENTS

4.1 General

MCC shall be manufacture or assembled in Indonesia. The MCCs shall be compartmented, ventilated, gasketed indoor type, and free standing metal enclosures. The metal thickness shall be according with technical specification E-280. They shall be sectionalized to contain bus and wiring, circuit breakers, starters, control circuits, devices and distribution panels as required.

Individual units shall consist of frames, mounting, disconnecting and wiring, molded case circuit breakers, magnetic contactors, overload relays, control circuit and terminal blocks.

The equipment shall comply with the recommendations of IEC 60439.

Provisions shall be made to padlock breakers or combination starters in the withdrawn position with a multiplicity of padlocks.

Equipment of the same type and rating on this Contract shall be physically and functionally interchangeable.

The main current carrying parts, insulators, supports and housings shall have sufficient mechanical strength to withstand, without incurring damage, the effect of any momentary current resulting from a three-phase, line-to-line, or line-to-ground short circuit. The current shall be rms value, including the direct current component, during the maximum cycle corresponding to the rated short circuit currents specified.

4.2 Structures

The rear section of each structure shall have ample space for bringing power and control cable directly to the terminal blocks at the motor control units.

All equipment shall be constructed of non-hygroscopic and non-flammable materials. The apertures for secondary bus wires shall be ‘bushed’ to prevent damage to wires from sharp edges of metal.

Anti condensation heaters shall be designed to operate from a 230 V AC supply. The heaters shall be sized such that overheating within the cubicle cannot occur with the heater continuously switched on. Each heater shall be controlled from a double-pole switch with a white lamp labeled “Cubicle Heaters On”. The circuit breaker and indicating lamp shall be provided on each item of equipment.

Padlocking facilities or other approved type of locking devices shall be provided to prevent unauthorized operation or access to the equipment.

All MCCs shall be fitted with an approved telephone jack-in socket which shall be wired to terminals for connecting to the maintenance telephone system described in the Technical Specification E-271, Communication System. Telephone jack-in sockets shall be fitted on the front left-hand side of all motor control centers.

Ventilating openings shall be provided. All openings on outdoor switchgear shall be covered with screens or the equivalent to prevent entrance of rodents, snakes and large insects. All openings on indoor switchgear shall be protected with drip-proof louvers. The switchgear shall have a degree of protection of at least IP 42 for indoor switchgear and IPW 54 for outdoor switchgear.

A copper ground bus shall be installed in each structure and bonded or bolted to all parts of the structure.

4.3 Buses

Each MCC shall be equipped with one (1) set of 3-phase and neutral main buses, a set of 3-phase and neutral auxiliary buses for each vertical multi-tier arrangement of units, and a ground bus.

The neutral bus shall be insulated from ground in a similar manner to the phase buses and shall be provided in addition to the MCC ground bus. Isolating links shall be provided in the neutral connection to each circuit.

Main buses, auxiliary buses, ground bus and connection between the several pieces of apparatus forming the equipment of a cubicle shall be of high conductivity copper in accordance with IEC 60028, and have a conductivity of not less than 99.9% of that standard.

Main buses shall be contained in a separate compartment within the MCC and access shall only be possible by means of bolted-on sheet steel covers. Main buses and bus connections shall not be exposed when covers and doors are opened for access to the remainder of the MCC.

Each phase conductor of the main buses, including all through joints and tapping connections shall, in addition to being spaced at such intervals to give the necessary air clearance for the voltage rating, be completely insulated or segregated with removable fire-resistant barriers in an approved manner.

All buses, connections and their supports shall be capable of carrying the short-time current associated with their short-circuit ratings for a period of one (1) second.

Bolted bus and bar joints and main and auxiliary contacts shall be silver plated. Main buses and connections shall be clearly marked to indicate the phase and shall be arranged in the order R, S, T from front to back, top to bottom or left to right as viewed from the switching device operating mechanism side.

4.4 Circuit Breakers

Circuit breakers shall be 400 V, 50 Hz, 3 poles with thermal magnetic trip units. Breakers shall be completely enclosed in a molded case.

Incoming feeder breakers to an MCC shall have combined time delay thermal trip protection and instantaneous magnetic trip protection in one (1) complete assembly.

The above protective actions shall be so coordinated that overcurrents will be cleared by the thermal action and short circuits will be cleared by the magnetic action.

Regardless of which tripping device serves to clear the circuit, all poles of the breaker shall simultaneously open automatically.

Circuit breakers used in combination starters for motor service shall have adjustable magnetic trip only, in place of the inverse time thermal overload and instantaneous overcurrent trips.

Circuit breakers shall have a manual operating handle on the front of the compartment door. The position of the handle shall indicate whether the breaker is open or closed. Provision shall be made for locking the breaker in the open position with two (2) or more padlocks. The circuit breaker and operating devices on the doors shall be arranged to prevent opening or closing the doors unless the circuit breaker and operating devices are in the "Off" position. A defeater shall be provided to permit opening of the door when the breaker is in the "On" position.

4.5 Motor Starter Units

4.5.1 General

Each motor starter unit shall be provided with a circuit breaker, control transformer, contactor and protection. The contactor shall be the normal means of stopping and starting the motor.

Circuit breakers shall be 400 V, 50 Hz, 3 poles with instantaneous magnetic only trip units. Overload relays shall use no compensated bimetal elements.

Interlocks shall be provided to ensure that the unit access door can only be opened when the associated circuit breaker is open and the circuit breaker cannot be closed in the service position until the access door is closed.

4.5.2 Contactors

Contactors shall be of the air break type fitted with arc shields and shall be electrically held in. Contactor of the mechanically latched type shall be furnished where required or advised by the Owner. Contactors shall comply with the recommendation of IEC 60947.

Motors controlled by mechanically latched contactors, if furnished, shall have time delayed undervoltage protection.

The operating coil voltage for electrically held in contactors shall be 110 V AC, and shall be obtained from control transformers mounted in each starter unit.

Contactors for use with motors shall be suitable for direct online starting and shall be capable of withstanding without damage the motor locked rotor current until the associated overload relay operates.

Contactors for reversing starters shall be both mechanically and electrically interlocked.

All contactors shall be capable of uninterrupted and intermittent duties as required by the operating conditions in accordance with IEC 60947 for the required duty. In addition, rocker type contactors shall be capable of a mechanical life duty of 500,000 load operations and block contactors of a mechanical life duty of 1,000,000 load operations.

4.5.3 Control Voltage Transformers

Each motor starter unit shall be equipped with individual control voltage transformers rated at 110 V AC on the secondary side. The primary windings have fuses on both sides and shall be solidly connected to the primary side of the contactor with fully insulated and shrouded connections. One side of the secondary shall be grounded via a removable link and the other side shall be provided with a suitably rated fuse.

4.6 Distribution Panels

The Contractor shall provide AC distribution panels to supply AC power to all items of plant which require AC power. Distribution panels shall be provided with top and bottom removable metal gland plates which shall be drilled on site to suit the cable glands or conduit. All top entry points shall be sealed to prevent the ingress of moisture. All gland plates shall be fitted with gaskets.

Distribution panels shall be fitted with a neutral bus having facilities for terminating as many neutral cables as single phase circuits in the panels.

Each distribution panel shall be fitted with an internal ground bus and a ground pad located in an accessible position on the outside of the panel. All metal parts of the panel except current carrying parts shall be bonded together and to the ground bus.

4.7 Automatic Transfer Switches

Each Essential Service MCC shall be provided with two (2) power feeds through an automatic transfer switch.

Automatic transfer switches shall be 400 V, 50 Hz, 3 poles with automatically and mechanically transfer facilities to an emergency power source upon failure of the normal source.

The transfer switches shall provide complete protection with field adjustable solid state voltage sensing logic to monitor each phase of the normal power supply.

The transfer switch shall be three pole double throw and shall transfer the load to the emergency source when the voltage of any phase drops to 70% of normal voltage. The time delay for transferring to the emergency source shall be adjustable between 0.5 and 6 seconds (factory set to 3 seconds). The load shall be transferred back to the normal source when the voltage of all three phases is restored to at least 90% of normal for a period adjustable from 0 to 30 minutes (factory set to 15 minutes). A test switch shall be provided to simulate failure of the normal source.

The transfer switch shall be operated by a momentarily energized single coil mechanism. The transfer switch shall be interlocked electrically and mechanically to prevent one (1) set of contacts from closing before the other opens. The power

used for transferring shall be obtained from the source to which the load is to be transferred.

The transfer switch shall be equipped with an undervoltage alarm set to operate when the voltage of the emergency source falls to 75% of normal. At least one (1) normally-open contact shall be provided rated 0.5 ampere at 250 volt DC inductive.

It shall not be possible for load circuits to be connected to normal and emergency sources simultaneously, regardless of whether the switch is electrically or manually operated. The switch shall have a provision for manual operation for load circuit maintenance. A transfer switch position indicator shall be visible from the front to show to which source the switch is connected.

The operating mechanism shall be designed such that it is impossible for the operating mechanism to stop between the normal and emergency positions. The mechanism shall be mechanically held on either source without the use of hooks. Delicate latches or spring operated switches are not acceptable. It shall have a minimum mechanical endurance of 100,000 operations.

The transfer switch shall be equipped with a mechanical operation counter.

4.8 Grounding

Grounding connections to the station grounding system and/or building steel shall be provided at each end of the bus. The ground bus dimensions shall be in no case be less than 7 x 50 mm copper or equivalent.

4.9 Control Devices and Wiring

All internal wiring shall be only connected to one side of the terminal blocks. Other side shall be used for the external connections.

All connections internal of MCCs, including circuit breakers, starters and miscellaneous equipment, shall be made with insulated lugs.

Wiring and schematic diagrams covering all electrical equipment that is factory wired as an integral part of this Technical Specification shall be provided.

4.10 Nameplates

Nameplates shall be provided for all MCCs. The nameplate shall list all MCC data required by IEC 60947.

All nameplates shall be in accordance with the Technical Specification E-280, unless otherwise specified herein.

Each MCC shall be provided with a main label and danger/caution labels where applicable. Main labels and caution labels shall be engraved in both the English and the Indonesian languages.

All components such as switches, circuit breakers, relays and other devices, whether mounted inside or on the surface of the compartments shall have identifying labels which shall be related to the identifying references on the arrangement drawing, schematic or wiring diagrams. Circuit identification labels shall be provided at the front of all units and an identical label shall be fitted to the front of all withdrawal moving portions. The wording shall be in the English language only.

Danger/caution labels shall have white letters on a red background and when danger or warning labels are fitted they shall not be mounted on removable parts. The size of danger labels shall be such that they are easily readable. The wording shall be in both the English and the Indonesian languages.

4.11 Tools

Four (4) test cabinets with control switch, indicating lights and control cable connecting devices shall be provided for field test and calibration.

5. PERFORMANCE TESTING AND GUARANTEES

5.1 General

The Contractor may submit test certificates of tests which have been carried out on identical equipment. Notwithstanding any provision in a standard, the Owner shall have the right to accept or reject such certificates in lieu of the specified tests.

The type tests prescribed shall be carried out at the Contractor's expense in all cases where such certificates are not available or rejected.

5.2 Shop Tests

Full tests shall be made by the manufacturer on the complete MCC equipment in accordance with the tests specified in the relevant IEC recommendations and ANSI C37.20 to determine the design performance and operating characteristics.

The tests shall include, but not be limited to, the followings:

- (1) Continuity and polarity tests on all coils and circuits.
- (2) Functional tests on all relays.
- (3) Check of electrical and mechanical interlocks.
- (4) Interchangeability of removable elements of corresponding rating such as circuit breakers, starters, contactors, etc.
- (5) Resistance measurement of the operating coils on each type of breaker.
- (6) Determination of minimum control voltage at which each breaker will trip or close successfully.
- (7) Check successful operation of trip and close mechanism at specified minimum and maximum control voltages.
- (8) Measurement of current at minimum and normal control voltage impressed on each operating coil.
- (9) Determination of speed of opening and closing operations of each size of circuit breaker, such results to be recorded and shall be proved satisfactory.

5.3 Site Tests

The Contractor shall be responsible for the proper installation, alignment, necessary adjustments, site tests, inspection and putting into commercial services of all the equipment provided under this Contract.

After installation on site all equipment, where applicable, shall be dielectrically tested and shall be operated to prove that the operating gear, starters, contactors, tripping devices, protective gear, and interlocking are satisfactory and primary and/or secondary injection tests shall be carried out on all protective equipment.

The following site tests shall be performed to ensure a satisfactory system. The tests shall include, but not be limited to, the followings:

- (1) Mechanical tests.
- (2) Point-to-point wiring continuity checks on all circuits to confirm the final connections agree with the panel wiring diagrams.
- (3) Energizing of all circuits.
- (4) Check operation of all relays, instruments, alarms and trips.
- (5) Adjustments of all relays in line with the settings to be agreed with the Owner.
- (6) Functional testing of all equipment.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control shall conform to the requirements included in Part 4, Subsection 2.7.

The quality assurance/quality control procedures shall be as specified. These shall include, but not be limited to, the followings:

- (1) Material Certification
- (2) Continuity and Wiring Tests
- (3) Short Circuit Tests
- (4) Insulation and High Voltage Tests
- (5) Mechanical Tests
- (6) Functional Tests
- (7) Packaging and Shipping
- (8) Welding
- (9) Cleaning and Painting

The quality assurance/quality control documentation as specified shall include, but not be limited to, material Certificates, Shop Test Report and All other documentation required by applicable codes and standards

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